Boosting Techniques - Theory Answers

Q1. What is Boosting in Machine Learning?

Boosting is an ensemble learning technique that combines multiple weak learners to create a strong predictive model. Each new model focuses on correcting the mistakes made by previous models, thereby improving overall accuracy.

Q2. Difference between AdaBoost and Gradient Boosting

AdaBoost adjusts the weights of incorrectly classified samples so that future models focus more on difficult cases. Gradient Boosting trains models sequentially by minimizing a loss function using gradient descent.

Q3. How does regularization help in XGBoost?

Regularization reduces overfitting by penalizing overly complex models. XGBoost uses L1 and L2 regularization to improve generalization and model stability.

Q4. Why is CatBoost efficient for handling categorical data?

CatBoost can directly process categorical features without extensive preprocessing. It uses ordered target statistics and special encoding techniques to reduce overfitting and improve accuracy.

Q5. Real-world applications where boosting is preferred over bagging

Boosting is preferred in fraud detection, credit scoring, customer churn prediction, recommendation systems, and medical diagnosis where achieving the highest possible predictive accuracy is important.

Q10. Loan Default Prediction Pipeline

Preprocess data by handling missing values and encoding categorical variables. Use CatBoost when many categorical features exist, XGBoost for maximum performance, and AdaBoost for simpler problems. Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV. Evaluate using Precision, Recall, F1-Score, ROC-AUC, and PR-AUC because the dataset is imbalanced. The business benefits through reduced credit risk, better customer screening, improved profitability, and more informed lending decisions.